



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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• Experiment No.: 6

1. Title:

Use string functions and regex for email validation, data extraction, and text analysis.

2. Software/Tools Required:

Visual Studio Code, Web Browser (Google Chrome / Mozilla Firefox), Node.js (for local server execution)

3. Theory:

String manipulation is a common requirement in software development, particularly for text processing and data cleaning. JavaScript provides a comprehensive suite of built-in string methods. Methods like `toUpperCase()` and `toLowerCase()` alter casing, while `trim()` removes whitespace. The `split()` method is exceptionally useful for dividing a string into an array of substrings based on a specified separator, facilitating operations like word counting or domain extraction.

For more complex pattern matching and text search operations, JavaScript utilizes **Regular Expressions (Regex)**. A regular expression is a sequence of characters that forms a search pattern. Regex is incredibly powerful for validating highly structured data formats.

For example, validating an email address requires checking for a specific sequence of alphanumeric characters, followed by an '@' symbol, a domain name, and a top-level domain suffix. Regex can define this precise pattern, allowing the program to strictly accept or reject user input. Furthermore, Regex can be combined with the string `match()` method to search through unstructured text and extract all instances that fit a certain pattern, such as pulling all numerical digits out of a paragraph.

4. Output:

 file:///C:/Users/arnav/Downloads/js/arnavkamdi_24070521158/expt_6/index.html

Email Validation & Data Extraction

Enter Email Address:

Validate Email

✓ Valid Email Format

Username extracted: arnav

Domain extracted: sitnagpur.siu.edu.in

Enter Text for Analysis:

Extract & Analyze

Original Length: 50 characters

Word Count: 6

Numbers Extracted: 6

Emails Extracted: None

Lowercase Version:
experiment 6 text analysis completed successfully.

5. Result/Conclusion:

The text analysis and validation application was successfully implemented. Regular expressions were rigorously applied to validate email formats and successfully extract all numerical data from a provided text block. String functions were effectively utilized to parse the text, count words, and extract the domain name from the validated email address.